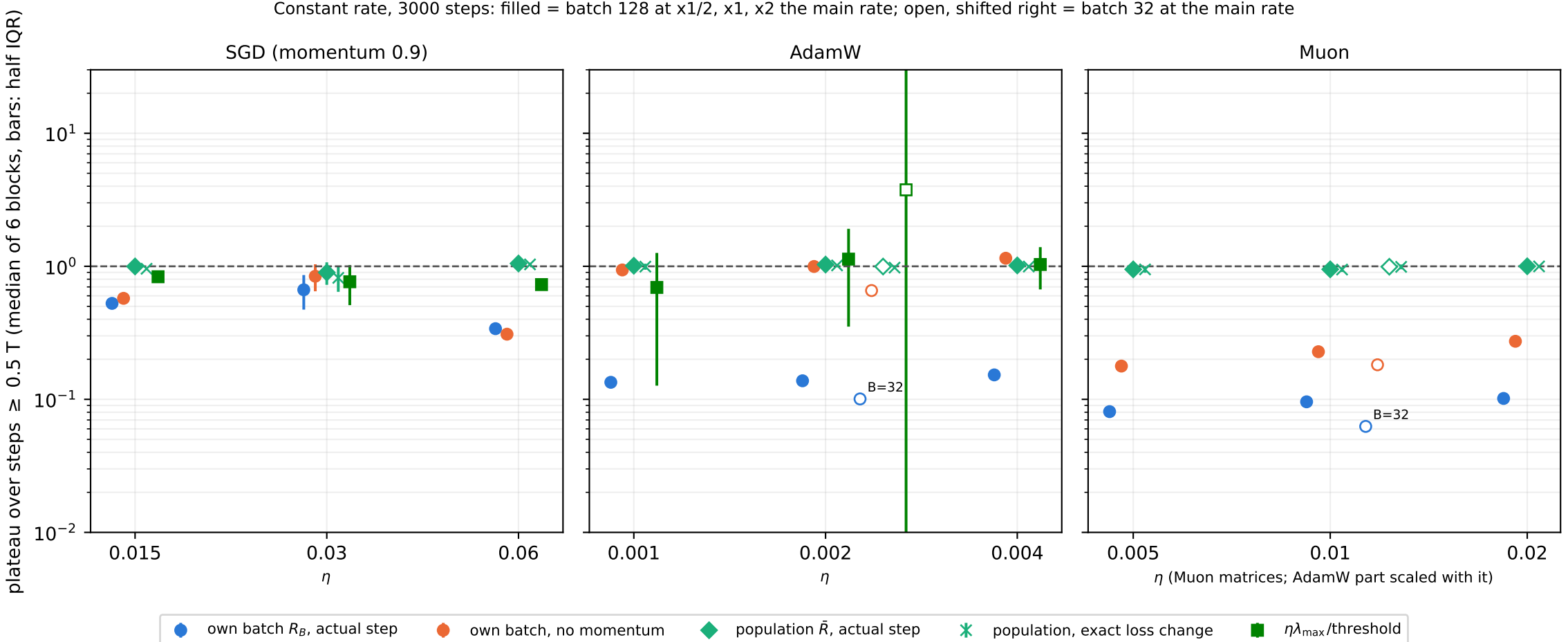
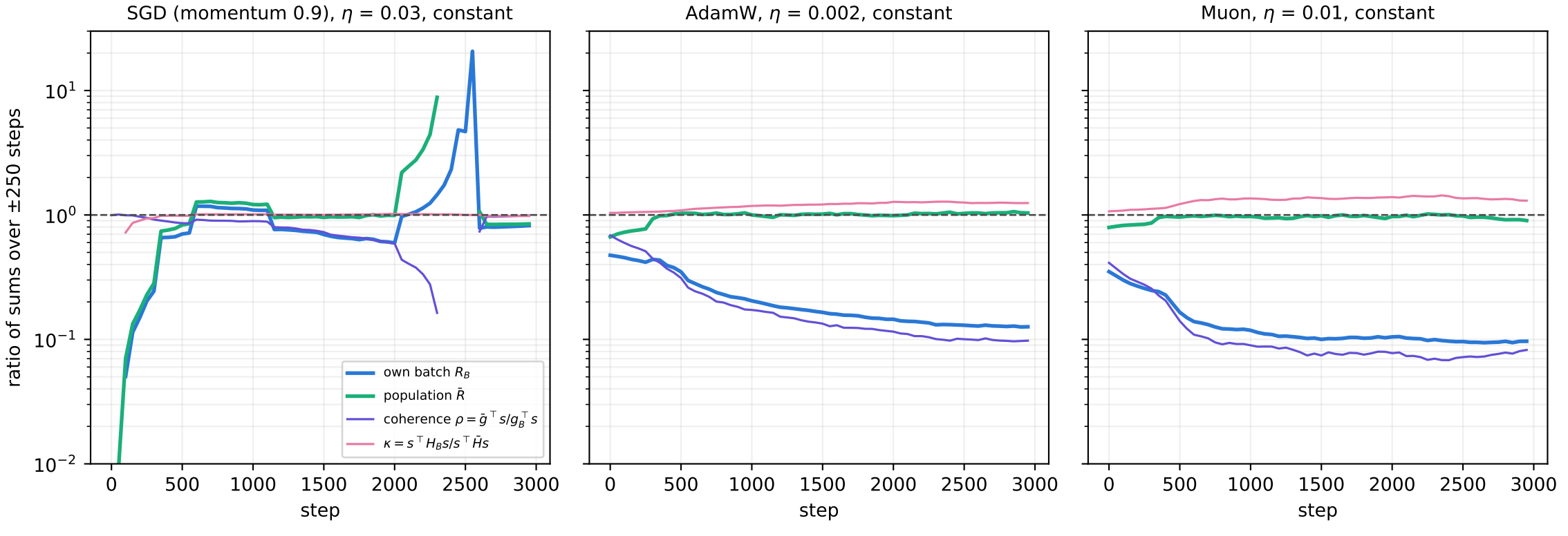
Stochastic edge of stability on TinyStories-10M (suite ts10m_stoch_eos, initialization and data order d0)

Does the curvature of each step's own minibatch along the update actually taken settle at the stability threshold?
With s the change of the parameters in one step (without decoupled weight decay) and g_B, H_B the gradient and
Hessian of the batch of that step, the step lowers the batch loss (to second order) iff
    R_B = s^T H_B s / (-2 g_B^T s) < 1   (s = -eta u: R_B = eta Q / 2, Q = u^T H_B u / g_B^T u).
Hessian-vector products on the step's batch every 10 steps (every step early on), and on 256 fresh training sequences
every 50 steps (population: R_pop = s^T H s / (-2 g^T s)). Also: the ratio from the exact one-step loss change, the
step without momentum (SGD: -eta g_B, Andreyev's eta * batch sharpness / 2; AdamW: -eta D^-1 g_B; Muon: -eta polar(g_B)
on its matrices), Muon's two parts, the step with weight decay, and eta lambda_max / threshold (Lanczos, 32 fixed seqs).
Runs: constant rate after 200 warmup steps, 3000 steps, batch 128, at x1/2, x1, x2 the main rate (SGD 0.03, momentum
0.9; AdamW 2e-3; Muon 0.01 with its AdamW part 2e-3, scaled together); batch 32 at the main rate; the main rate with
warmup and cosine decay over 10k steps. Plateau: median over 6 blocks of the second half (ratio of sums per block).

Plateaus, range over x1/2, x1, x2 (batch 128, constant rate):
  own batch, actual step   R_B   SGD 0.34-0.67, AdamW 0.13-0.15, Muon 0.08-0.10
  population, actual step   R_pop   SGD 0.90-1.05, AdamW 1.01-1.03, Muon 0.95-1.00
      from the exact loss change   SGD 0.82-1.03, AdamW 0.99-1.02, Muon 0.95-1.00
  own batch, no momentum   SGD 0.31-0.84, AdamW 0.94-1.15, Muon 0.18-0.27
  coherence rho = g^T s / g_B^T s   SGD 0.30-0.66, AdamW 0.10-0.12, Muon 0.06-0.07
  kappa = s^T H_B s / s^T H s   SGD 1.01-1.05, AdamW 1.22-1.26, Muon 1.37-1.38
  eta lambda_max / threshold   SGD 0.73-0.83, AdamW 0.69-1.13, Muon -
Batch 32:   R_B SGD -, AdamW 0.10, Muon 0.06; R_pop SGD -, AdamW 1.00, Muon 0.99
Cosine, 2nd half (eta falls to 10%): R_B SGD 0.20, AdamW 0.04, Muon 0.02; R_pop SGD 1.05, AdamW 0.95, Muon 0.91

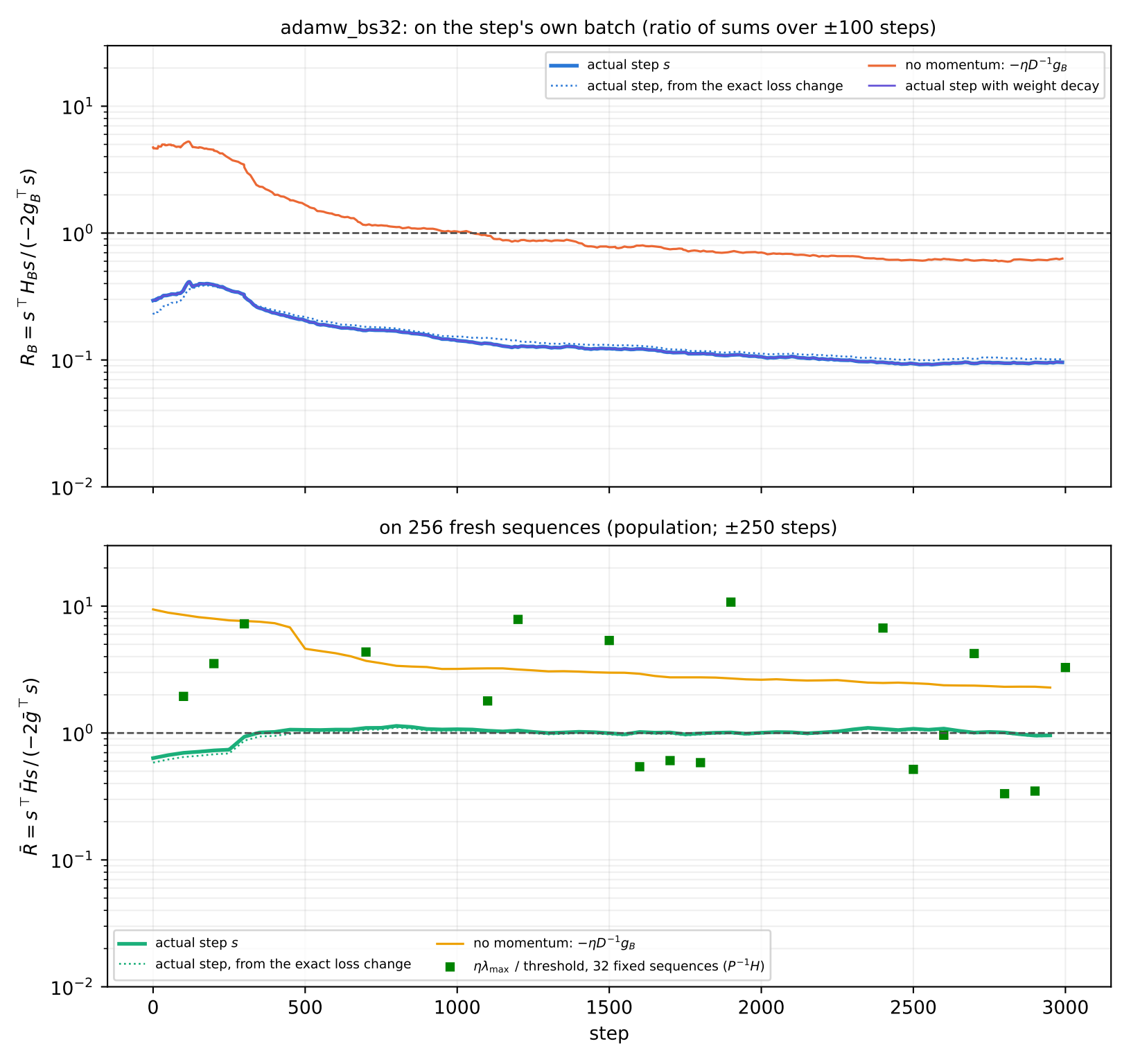
Findings
1. R_B does not settle at 1 for any optimizer: it falls throughout training, and it is far below 1 for AdamW and Muon.
2. R_pop, the same ratio with the population gradient and Hessian along the actual step, settles at 1 for all three
   optimizers, at every rate and batch size, within ~500 steps, and stays there while the cosine schedule lowers eta
   (when eta lambda_max / threshold falls well below 1). SGD at 0.03 blows up at step 2300 (loss 2.6 -> 6.5); R_pop
   rises above 1 in the 300 steps before. Muon: neither part sits at 1 alone (population: matrices ~0.84, AdamW part
   ~0.45); the cross term between them makes the full step's ratio 1.
3. R_B = R_pop * kappa * rho exactly. kappa ~ 1-1.4 (2.1 for Muon at batch 32), so R_B ~ rho: the own-batch ratio is
   the population threshold times the step's batch-population coherence (cf. the 2 rho_b / eta of 2609.34915), which
   falls as gradient noise dominates (smaller at batch 32, smaller as training goes on).
4. Along the literature directions on the own batch (no momentum) the ratio drifts too (AdamW along D^-1 g_B: from ~8
   down through 1 at constant rate, to 0.2 on the cosine schedule); it is not a plateau.
5. Exactly, 1 - R_pop(exact) = dL_pop / g^T S: R_pop ~ 1 means each step realizes only a few percent of its first-order
   decrease of the population loss. A stable step has R ~ eta lambda / 2 << 1, as early in training here.

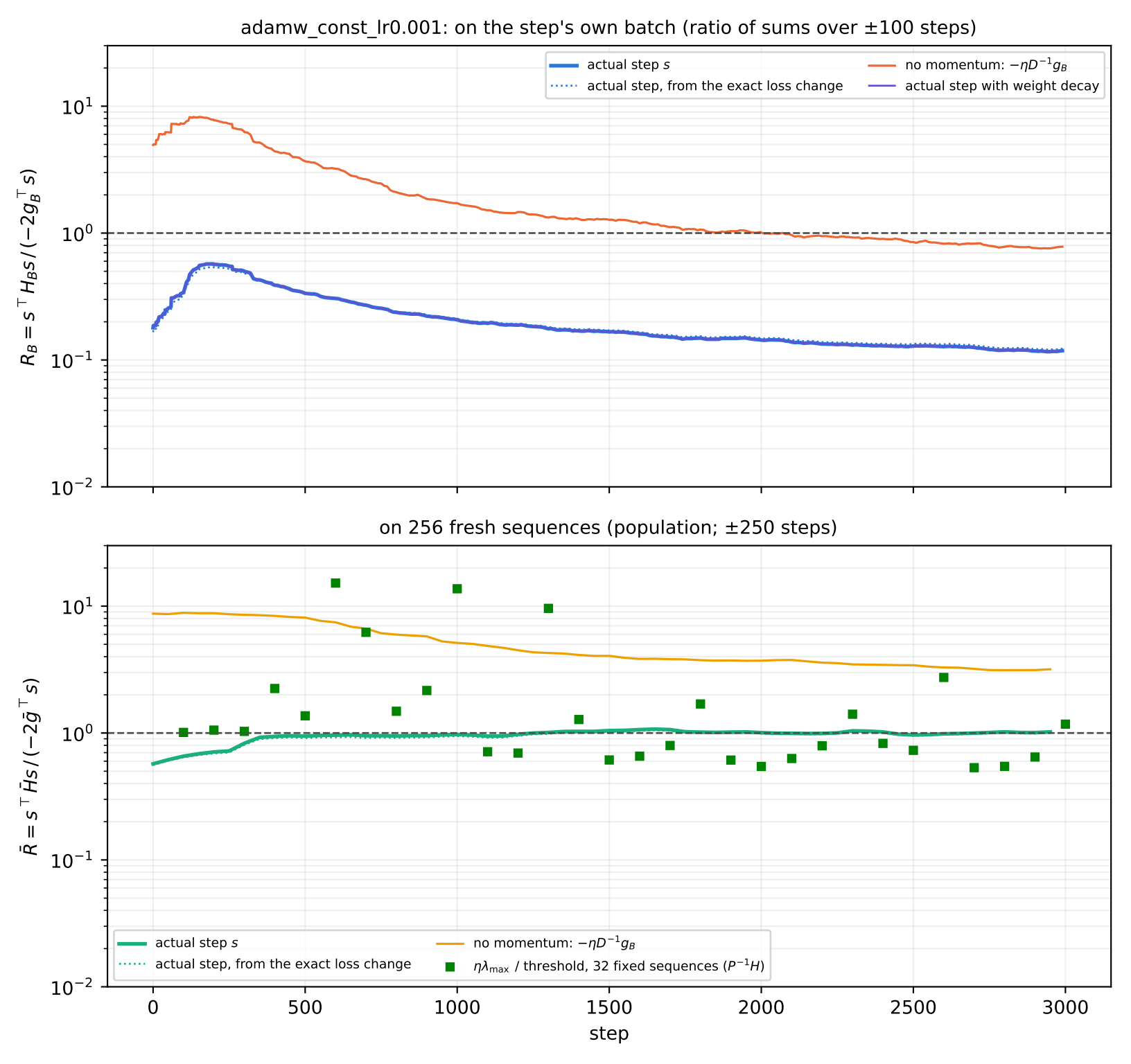

Constant rate, 3000 steps: filled = batch 128 at x1/2, x1, x2 the main rate; open, shifted right = batch 32 at the main rate
SGD (momentum 0.9)
AdamW
Muon
plateau over steps ≥ 0.5 T (median of 6 blocks, bars: half IQR)
10^{1}
10^{0}
10^{-1}
10^{-2}
0.015
0.03
0.06
0.001
0.002
0.004
0.005
0.01
0.02
B=32
B=32
η
η
η (Muon matrices; AdamW part scaled with it)
own batch R_B, actual step
own batch, no momentum
population $\bar{R}$, actual step
population, exact loss change
$\eta\lambda_{\max}$/threshold

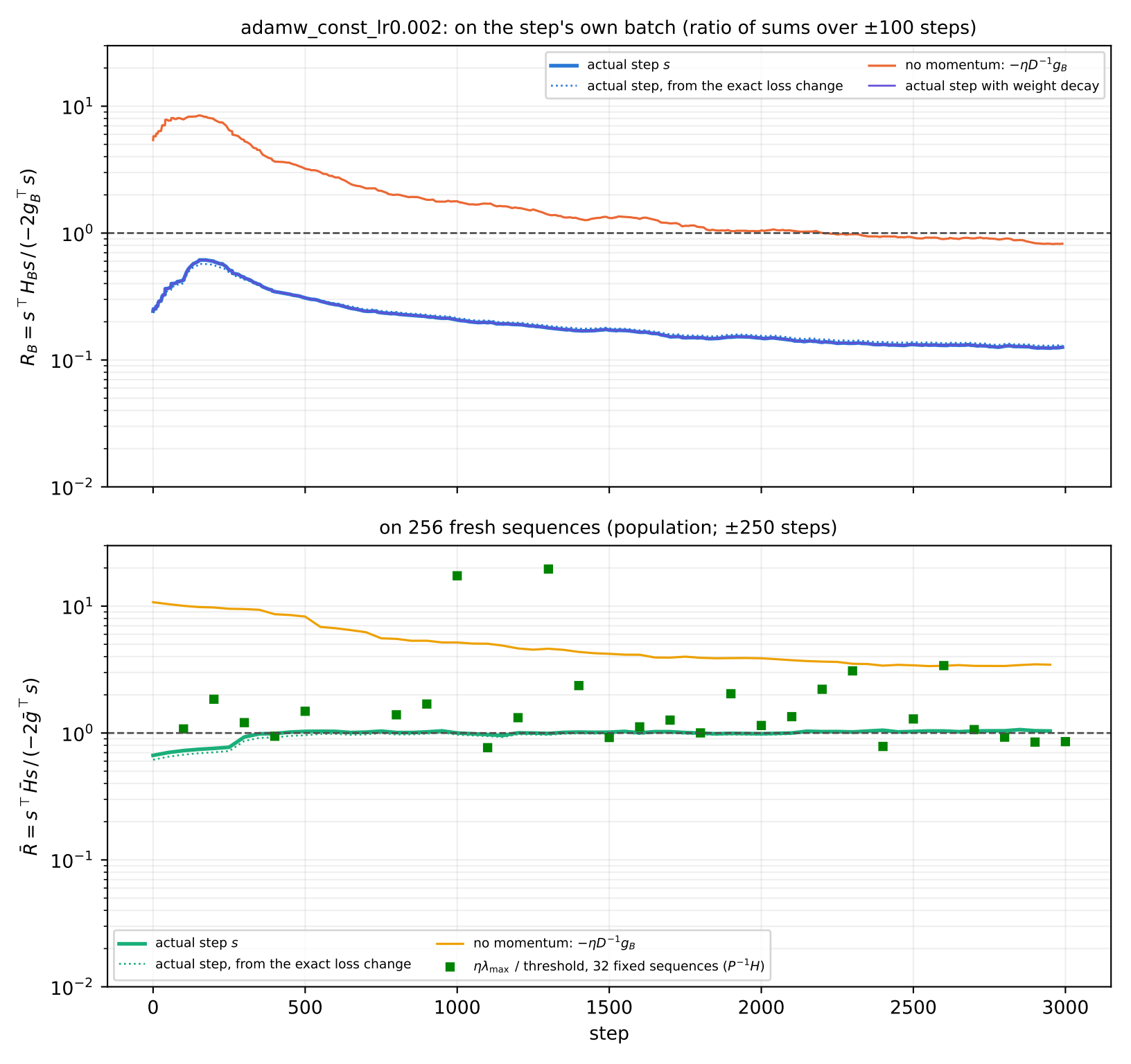

Why the own-batch ratio is not 1: $R_B = \bar{R}\,\kappa\,\rho$ exactly; $\bar{R} \approx 1$, $\kappa \approx$ 1-1.4, so $R_B \approx \rho$
SGD (momentum 0.9), $\eta = 0.03$, constant
AdamW, $\eta = 0.002$, constant
Muon, $\eta = 0.01$, constant
ratio of sums over ±250 steps
10^{1}
10^{0}
10^{-1}
10^{-2}
0
500
1000
1500
2000
2500
3000
step
own batch R_B
population $\bar{R}$
coherence $\rho = \bar{g}^{\top} s / g_B^{\top} s$
$\kappa = s^{\top} H_B s / s^{\top} \bar{H} s$

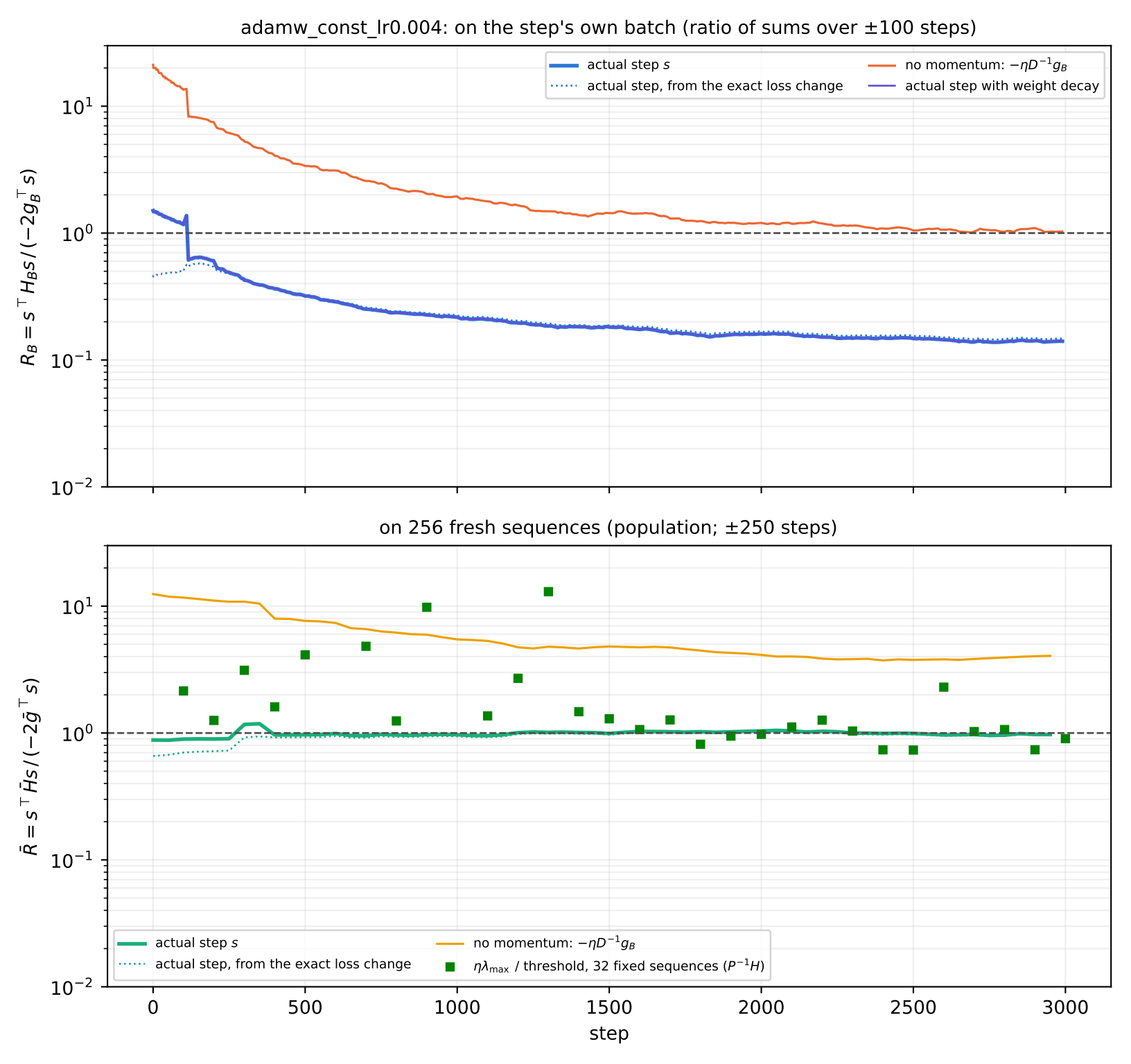
Plateaus over the second half of each run: median over 6 blocks of the ratio of sums in each block, ± half the IQR
B = own batch, pop = fresh sequences, mix = own-batch Hessian with the population gradient
rho = g_pop^T s / g_B^T s, kappa = s^T H_B s / s^T H_pop s, so that B.s = pop.s * kappa * rho
lam = eta lambda_max / threshold on the 32 fixed sequences (median)

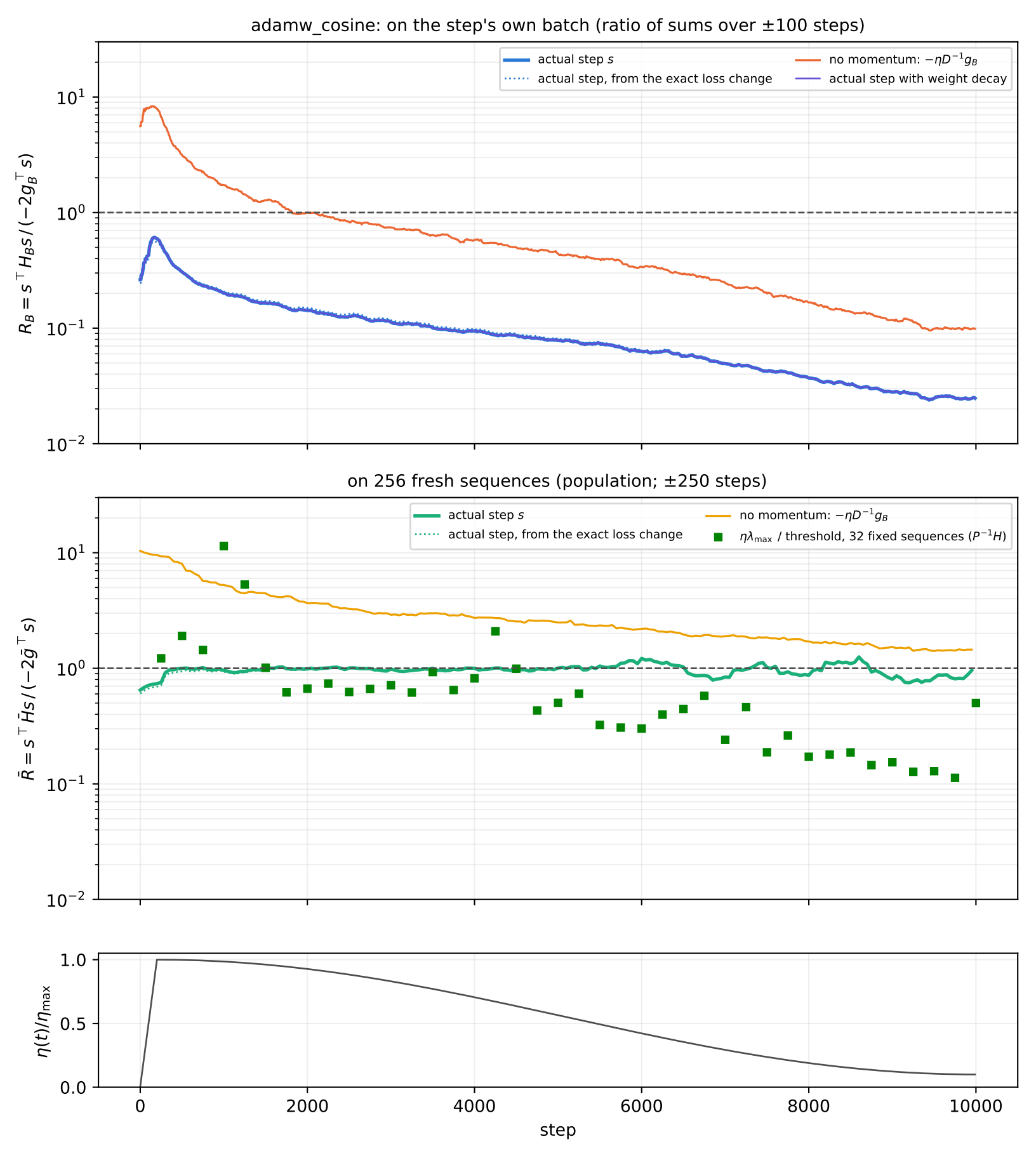
run	B.s	B.exact	B.nm	B.s.muon	B.s.adam	B.s+wd	pop.s	pop.exact	pop.nm	mix.s	rho	kappa	lam
adamw_bs32	0.101±0.01	0.108±0.01	0.657±0.04			0.101±0.01	0.996±0.03	0.977±0.03	2.527±0.12	1.503±0.02	0.071±0.01	1.484±0.06	3.757±164.11
adamw_const_lr0.001	0.134±0.01	0.139±0.01	0.939±0.09			0.134±0.01	1.010±0.05	0.994±0.05	3.591±0.26	1.253±0.06	0.107±0.00	1.216±0.01	0.694±0.57
adamw_const_lr0.002	0.138±0.01	0.144±0.01	0.997±0.07			0.138±0.01	1.031±0.02	1.015±0.02	3.647±0.25	1.303±0.03	0.104±0.01	1.261±0.01	1.132±0.78
adamw_const_lr0.004	0.153±0.01	0.160±0.01	1.150±0.07			0.153±0.01	1.015±0.03	0.996±0.03	3.929±0.16	1.283±0.06	0.118±0.00	1.264±0.02	1.032±0.36
adamw_cosine	0.044±0.01	0.045±0.01	0.209±0.08			0.044±0.01	0.951±0.06	0.951±0.06	1.803±0.21	1.140±0.08	0.039±0.01	1.222±0.02	0.262±0.16
muon_bs32	0.063±0.00	0.070±0.00	0.182±0.01	0.050±0.00	0.023±0.00		0.992±0.04	0.990±0.04	0.978±0.05	2.097±0.16	0.030±0.00	2.124±0.12	
muon_const_lr0.005	0.081±0.00	0.085±0.00	0.178±0.01	0.074±0.00	0.022±0.00		0.948±0.05	0.948±0.05	1.170±0.02	1.366±0.09	0.061±0.00	1.383±0.03	
muon_const_lr0.01	0.096±0.01	0.102±0.01	0.228±0.02	0.087±0.00	0.029±0.00		0.949±0.04	0.945±0.04	1.365±0.03	1.328±0.07	0.074±0.00	1.370±0.02	
muon_const_lr0.02	0.102±0.01	0.109±0.01	0.273±0.01	0.089±0.01	0.032±0.00		1.001±0.03	0.998±0.02	1.570±0.04	1.396±0.05	0.074±0.01	1.371±0.03	
muon_cosine	0.023±0.01	0.024±0.01	0.047±0.02	0.020±0.01	0.009±0.00		0.915±0.08	0.910±0.10	0.724±0.09	1.235±0.11	0.019±0.01	1.337±0.03	
sgd_const_lr0.015	0.526±0.02	0.516±0.02	0.574±0.04				0.998±0.02	0.958±0.02	0.829±0.03	1.024±0.02	0.518±0.02	1.028±0.00	0.835±0.06
sgd_const_lr0.03	0.666±0.19	0.643±0.13	0.842±0.19				0.898±0.17	0.818±0.18	0.967±0.16	0.890±0.18	0.662±0.17	1.005±0.01	0.765±0.25
sgd_const_lr0.06	0.340±0.02	0.338±0.02	0.309±0.03				1.050±0.03	1.033±0.03	0.609±0.04	1.100±0.03	0.300±0.03	1.049±0.00	0.729±0.04
sgd_cosine	0.202±0.05	0.202±0.05	0.109±0.05				1.051±0.03	1.041±0.03	0.319±0.06	1.066±0.03	0.185±0.05	1.015±0.00	0.288±0.13

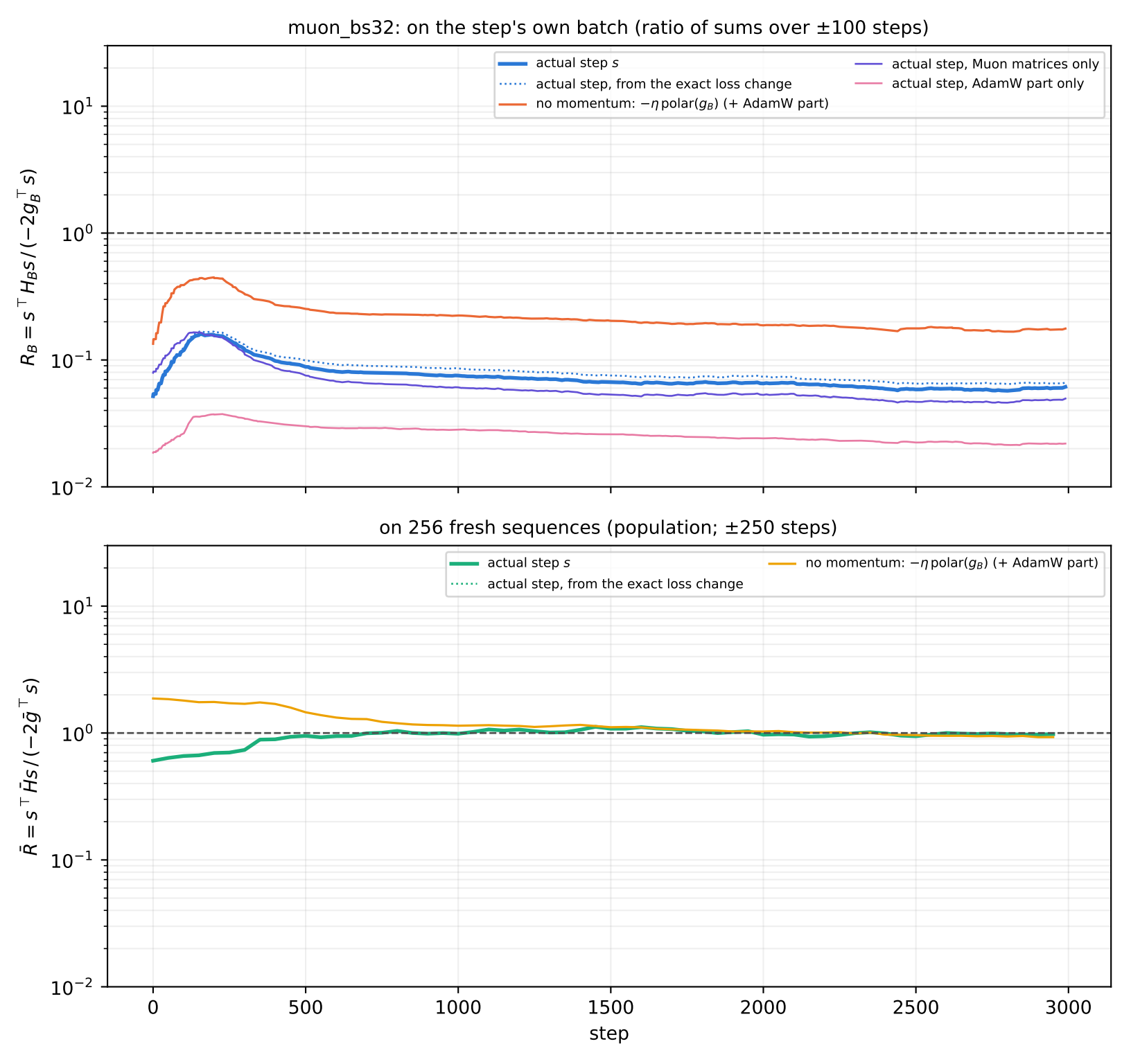

adamw_bs32: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta D^{-1} g_B$
actual step with weight decay
$R_B = s^\top H_B s / (-2 g_B^\top s)$
10^1
10^0
10^{-1}
10^{-2}
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta D^{-1} g_B$
$\eta\lambda_{\max}$ / threshold, 32 fixed sequences ($P^{-1}H$)
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
0
500
1000
1500
2000
2500
3000
step

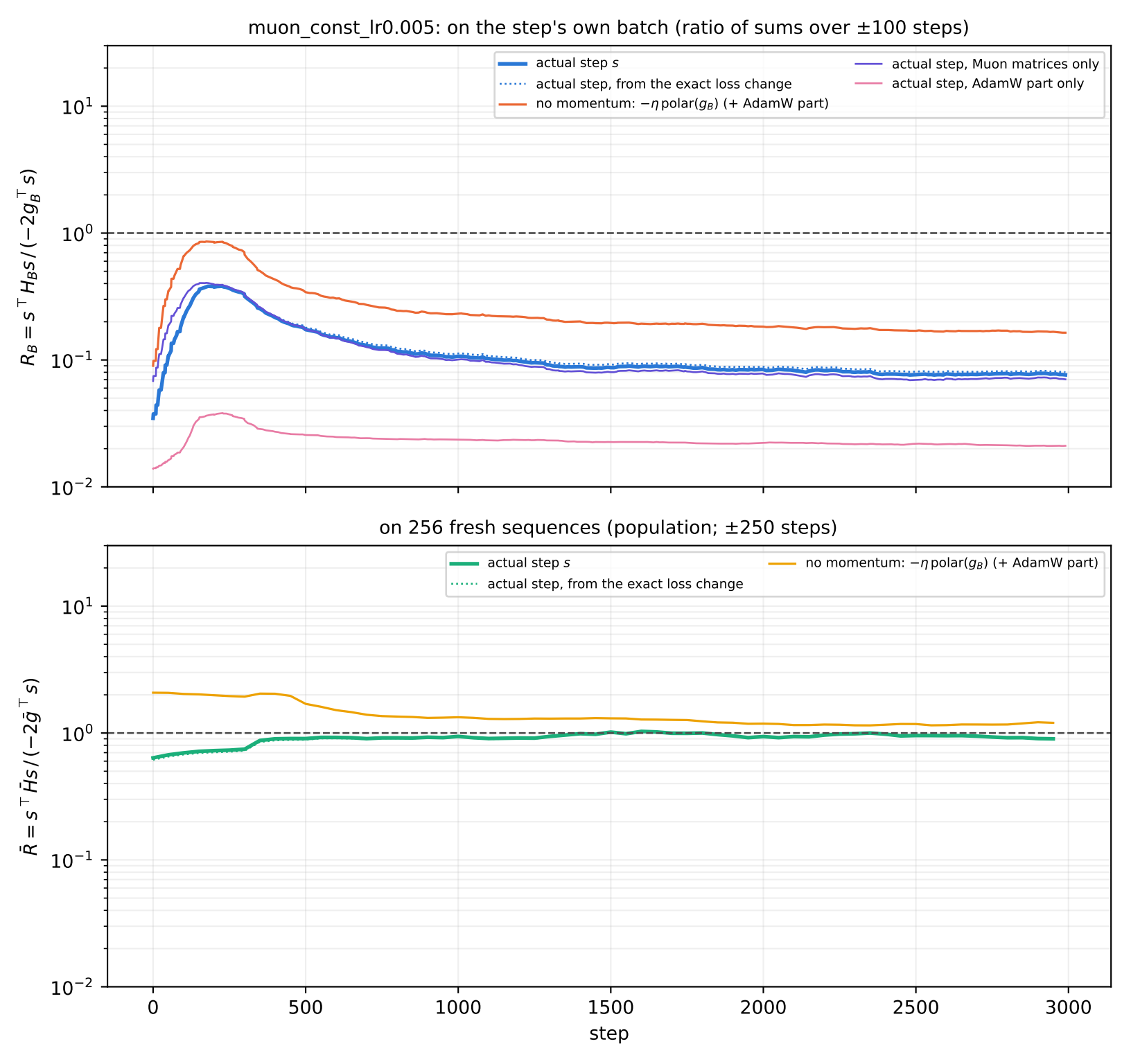

adamw_const_lr0.001: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta D^{-1} g_B$
actual step with weight decay
$R_B = s^\top H_B s / (-2 g_B^\top s)$
10^1
10^0
10^{-1}
10^{-2}
on 256 fresh sequences (population; ±250 steps)
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
actual step s
actual step, from the exact loss change
no momentum: $-\eta D^{-1} g_B$
$\eta\lambda_{\max}$ / threshold, 32 fixed sequences ($P^{-1}H$)
0
500
1000
1500
2000
2500
3000
step

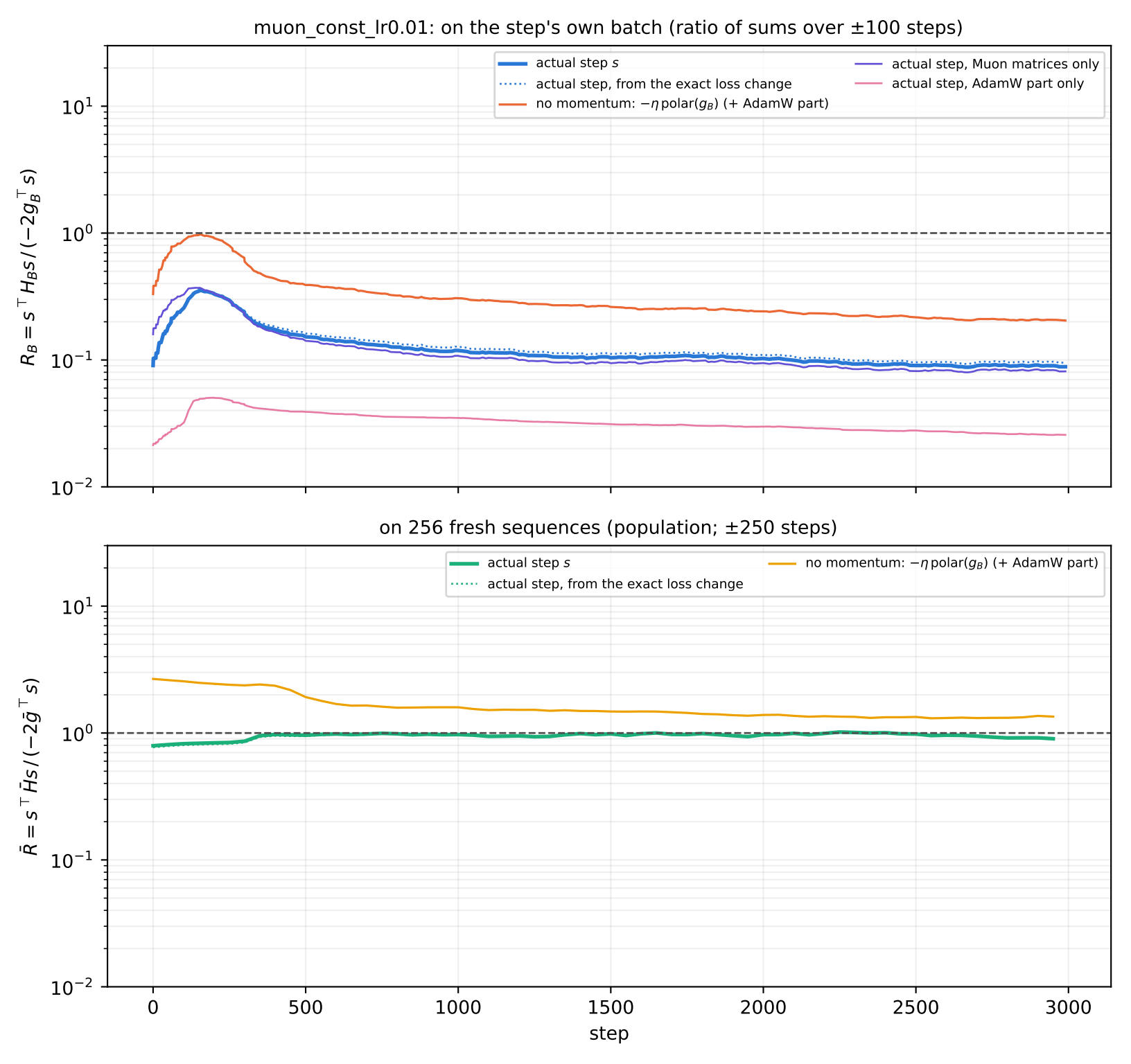

adamw_const_lr0.002: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta D^{-1} g_B$
actual step with weight decay
$R_B = s^\top H_B s / (-2 g_B^\top s)$
10^1
10^0
10^{-1}
10^{-2}
on 256 fresh sequences (population; ±250 steps)
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
actual step s
actual step, from the exact loss change
no momentum: $-\eta D^{-1} g_B$
$\eta\lambda_{\max}$ / threshold, 32 fixed sequences ($P^{-1}H$)
0
500
1000
1500
2000
2500
3000
step

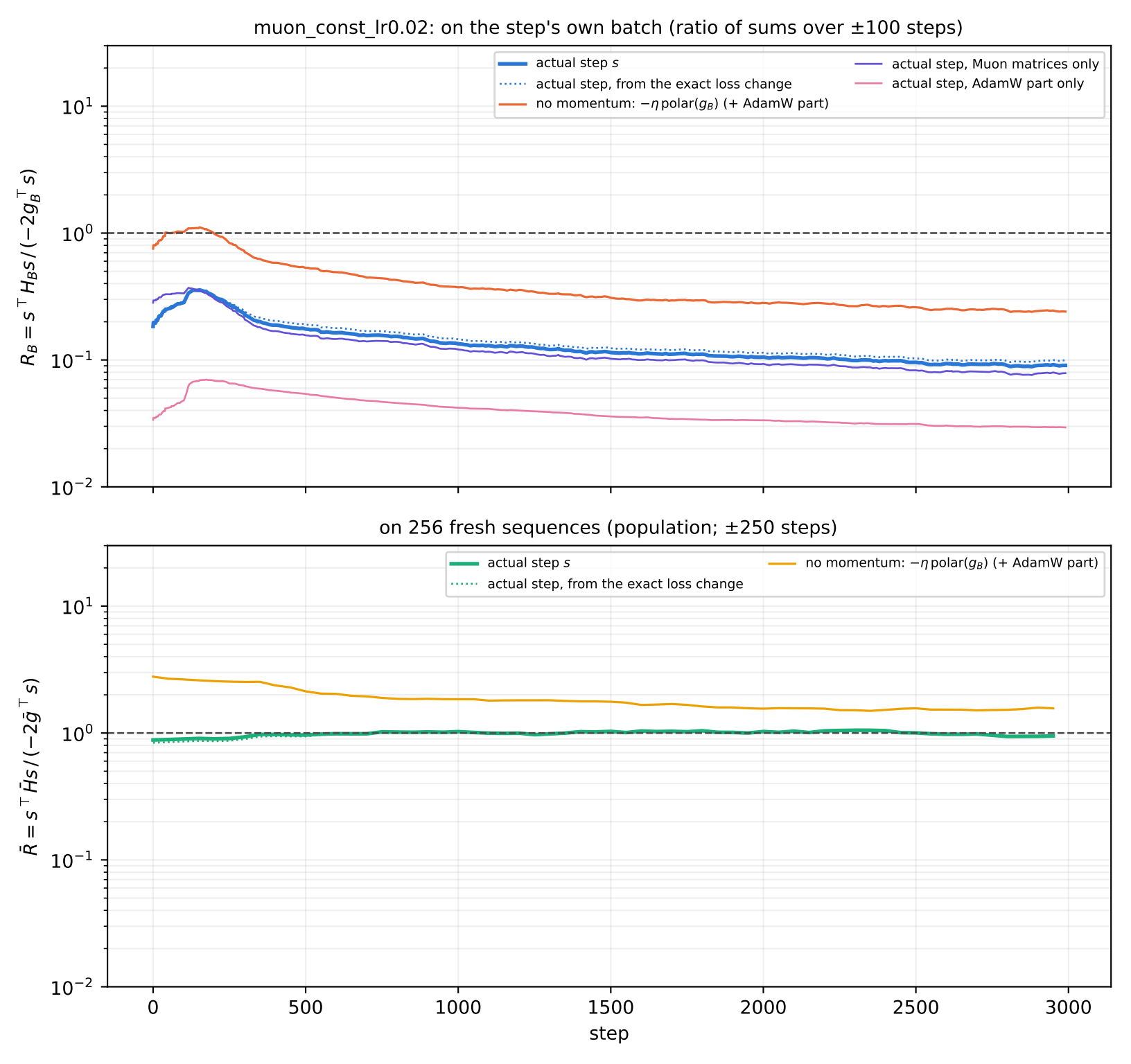

adamw_const_lr0.004: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta D^{-1} g_B$
actual step with weight decay
$R_B = s^\top H_B s / (-2 g_B^\top s)$
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta D^{-1} g_B$
$\eta\lambda_{\max}$ / threshold, 32 fixed sequences ($P^{-1}H$)
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
step

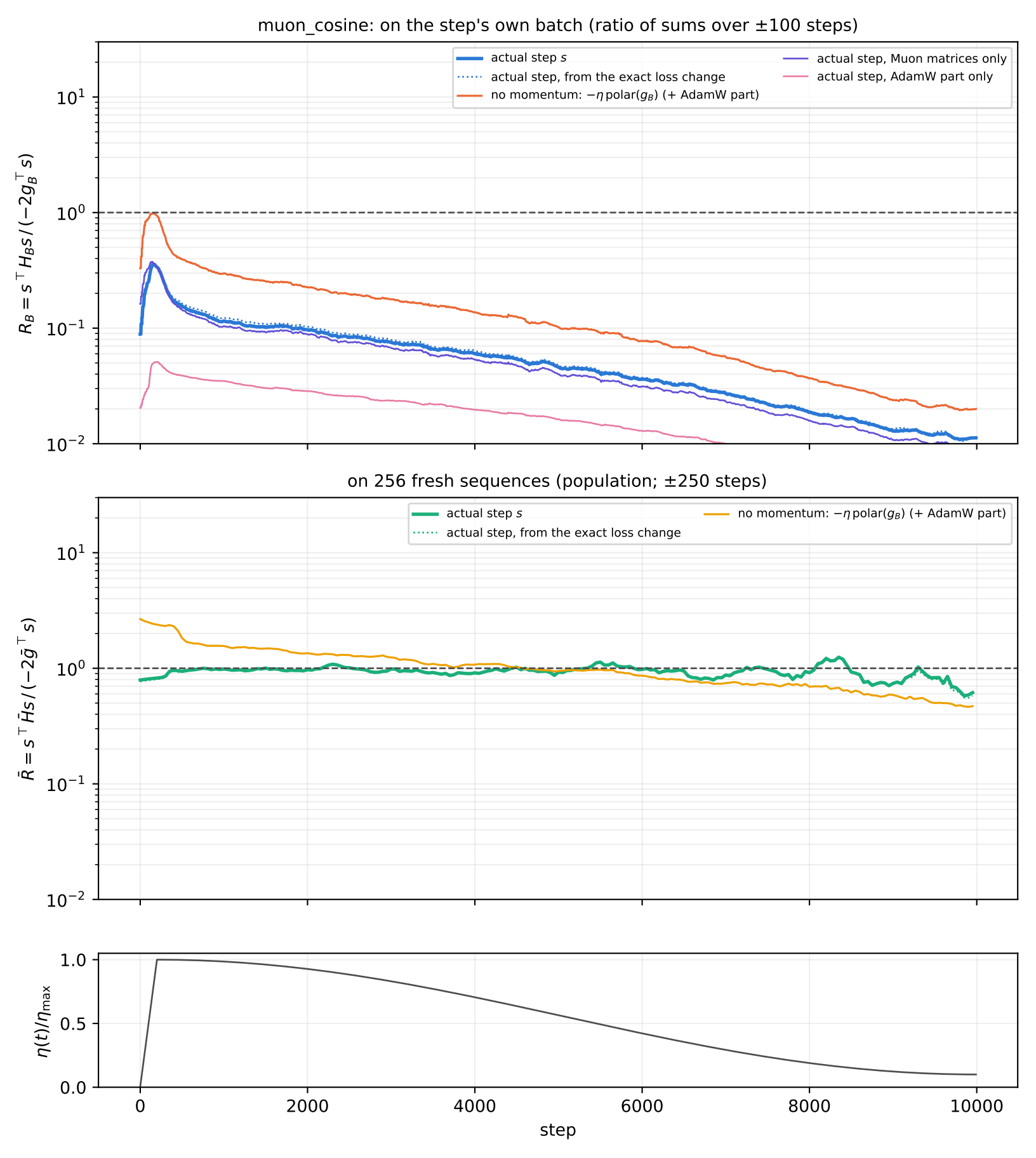

adamw_cosine: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta D^{-1} g_B$
actual step with weight decay
$R_B = s^\top H_B s / (-2 g_B^\top s)$
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta D^{-1} g_B$
$\eta\lambda_{\max}$ / threshold, 32 fixed sequences ($P^{-1}H$)
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
$\eta(t)/\eta_{\max}$
step

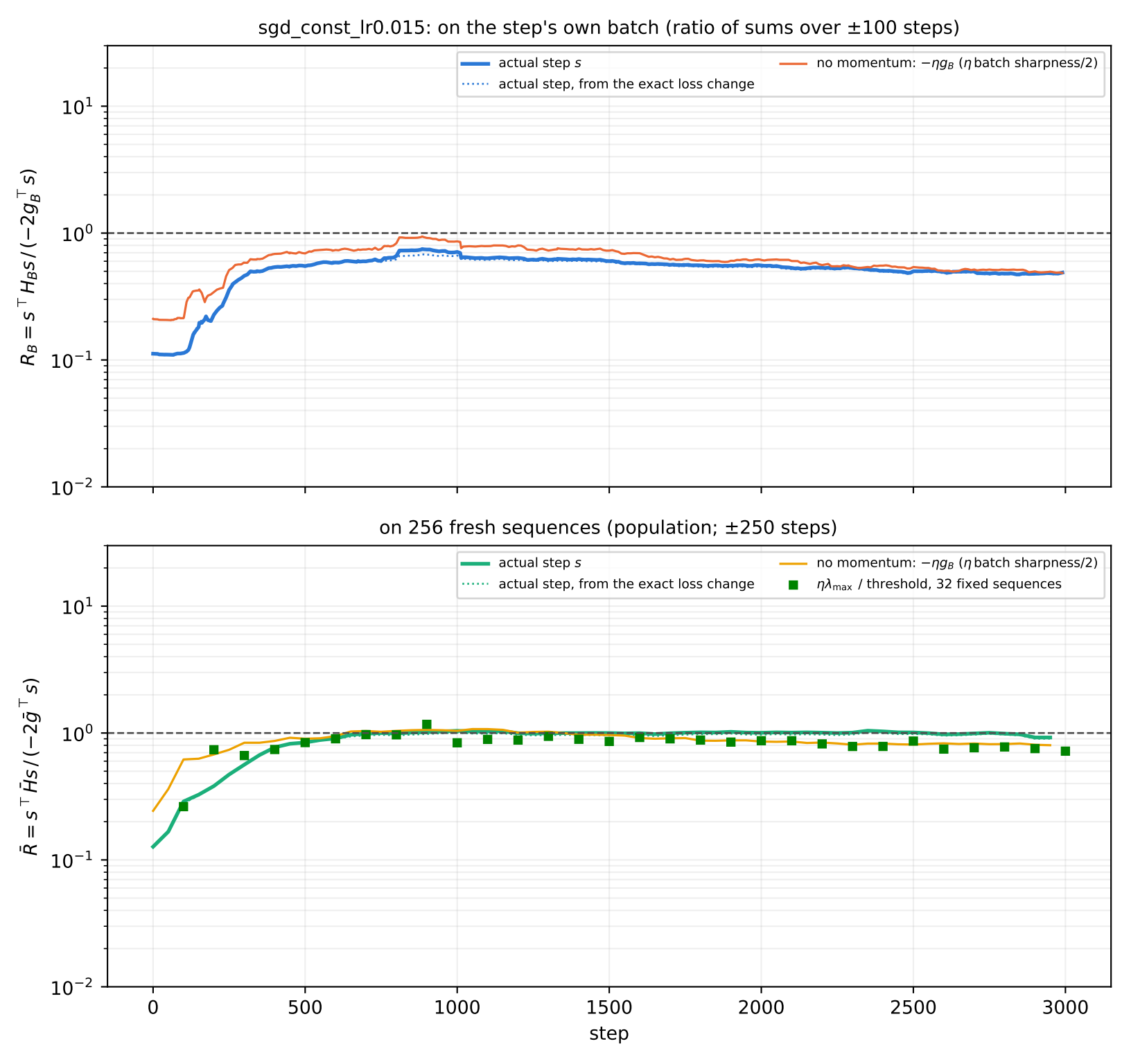

muon_bs32: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta\,\mathrm{polar}(g_B)$ (+ AdamW part)
actual step, Muon matrices only
actual step, AdamW part only
$R_B = s^\top H_B s / (-2 g_B^\top s)$
10^1
10^0
10^{-1}
10^{-2}
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta\,\mathrm{polar}(g_B)$ (+ AdamW part)
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
10^1
10^0
10^{-1}
10^{-2}
0
500
1000
1500
2000
2500
3000
step


muon_const_lr0.005: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta\,\mathrm{polar}(g_B)$ (+ AdamW part)
actual step, Muon matrices only
actual step, AdamW part only
$R_B = s^\top H_B s / (-2 g_B^\top s)$
10^1
10^0
10^{-1}
10^{-2}
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta\,\mathrm{polar}(g_B)$ (+ AdamW part)
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
10^1
10^0
10^{-1}
10^{-2}
0
500
1000
1500
2000
2500
3000
step

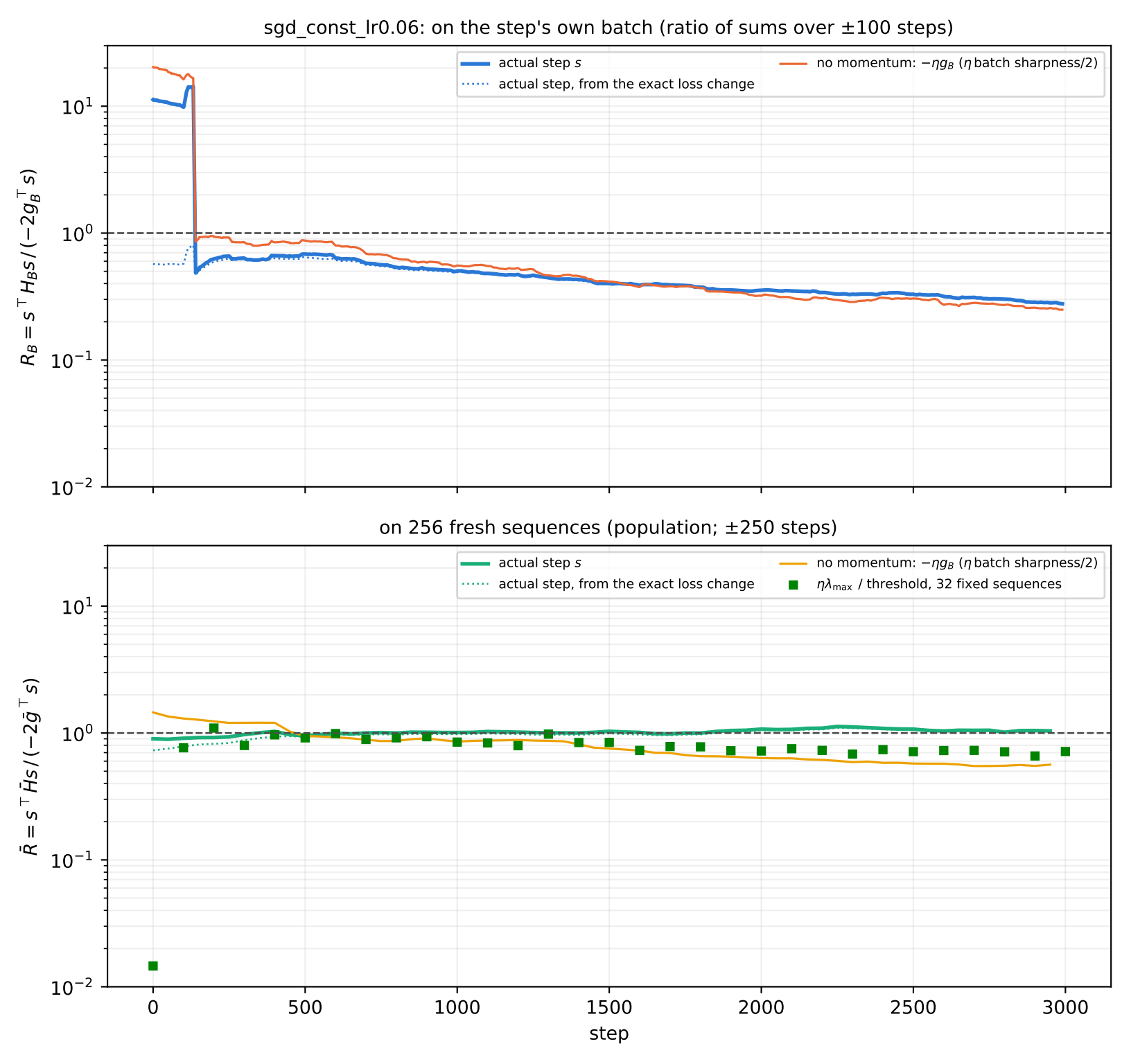

muon_const_lr0.01: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta\,\mathrm{polar}(g_B)$ (+ AdamW part)
actual step, Muon matrices only
actual step, AdamW part only
$R_B = s^\top H_B s / (-2 g_B^\top s)$
10^1
10^0
10^{-1}
10^{-2}
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta\,\mathrm{polar}(g_B)$ (+ AdamW part)
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
10^1
10^0
10^{-1}
10^{-2}
0
500
1000
1500
2000
2500
3000
step

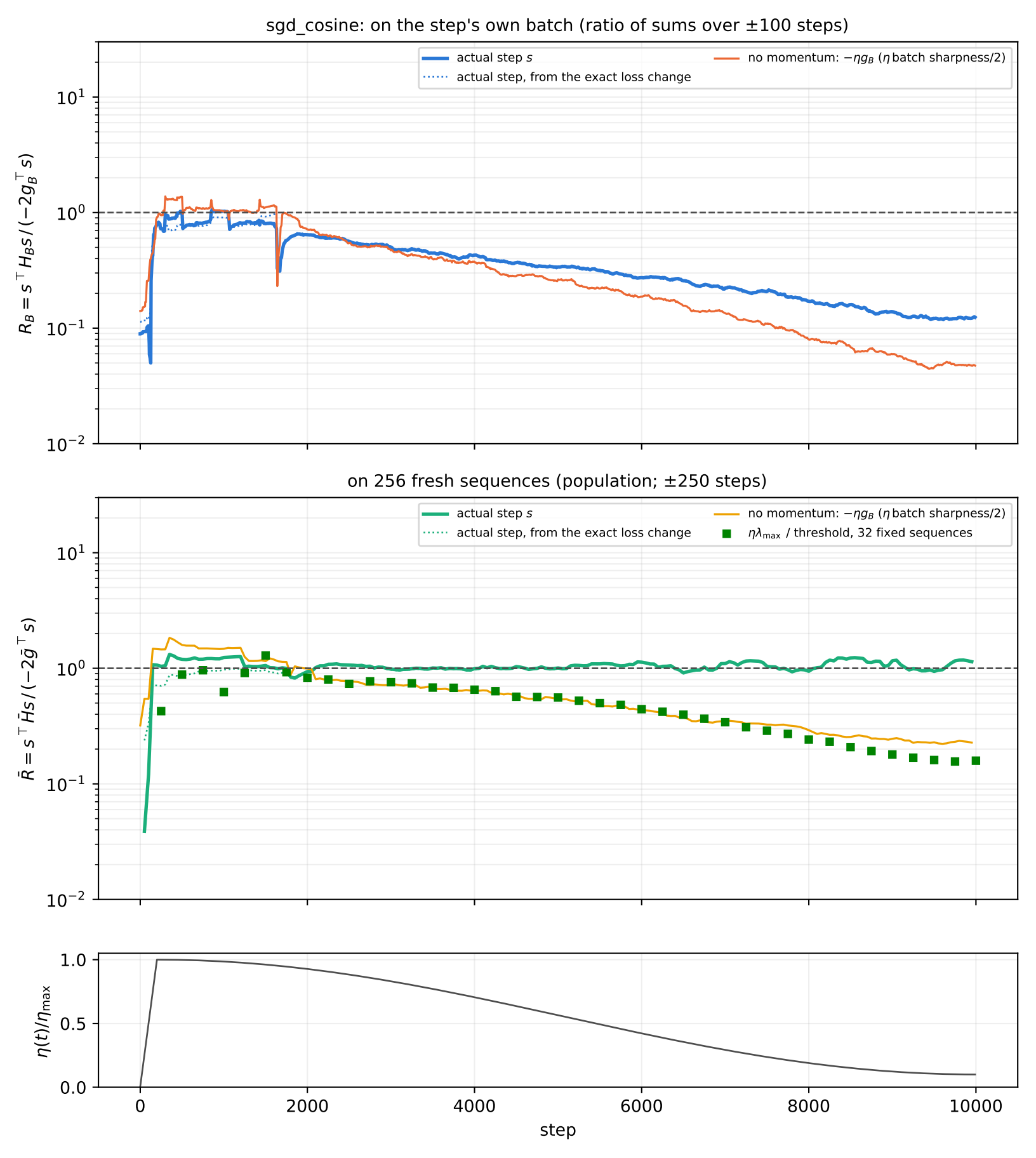

muon_const_lr0.02: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta\,\mathrm{polar}(g_B)$ (+ AdamW part)
actual step, Muon matrices only
actual step, AdamW part only
$R_B = s^\top H_B s / (-2 g_B^\top s)$
10^1
10^0
10^{-1}
10^{-2}
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta\,\mathrm{polar}(g_B)$ (+ AdamW part)
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
10^1
10^0
10^{-1}
10^{-2}
0
500
1000
1500
2000
2500
3000
step


muon_cosine: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta\,\mathrm{polar}(g_B)$ (+ AdamW part)
actual step, Muon matrices only
actual step, AdamW part only
$R_B = s^\top H_B s / (-2 g_B^\top s)$
10^1
10^0
10^{-1}
10^{-2}
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta\,\mathrm{polar}(g_B)$ (+ AdamW part)
$\tilde{R} = s^\top \tilde{H} s / (-2 \tilde{g}^\top s)$
10^1
10^0
10^{-1}
10^{-2}
$\eta(t)/\eta_{\max}$
1.0
0.5
0.0
0
2000
4000
6000
8000
10000
step


sgd_const_lr0.015: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta g_B$ (η batch sharpness/2)
$R_B = s^\top H_B s / (-2 g_B^\top s)$
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta g_B$ (η batch sharpness/2)
$\eta\lambda_{\max}$ / threshold, 32 fixed sequences
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
step


sgd_const_lr0.03: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta g_B$ (η batch sharpness/2)
$R_B = s^\top H_B s / (-2 g_B^\top s)$
10^1
10^0
10^{-1}
10^{-2}
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta g_B$ (η batch sharpness/2)
$\eta\lambda_{\max}$ / threshold, 32 fixed sequences
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
10^1
10^0
10^{-1}
10^{-2}
0
500
1000
1500
2000
2500
3000
step


sgd_const_lr0.06: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta g_B$ (η batch sharpness/2)
$R_B = s^\top H_B s / (-2 g_B^\top s)$
10^1
10^0
10^{-1}
10^{-2}
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta g_B$ (η batch sharpness/2)
$\eta\lambda_{\max}$ / threshold, 32 fixed sequences
$\bar{R} = s^\top \bar{H} s / (-2 \bar{g}^\top s)$
10^1
10^0
10^{-1}
10^{-2}
0
500
1000
1500
2000
2500
3000
step


sgd_cosine: on the step's own batch (ratio of sums over ±100 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta g_B$ (η batch sharpness/2)
$R_B = s^\top H_B s / (-2 g_B^\top s)$
10^1
10^0
10^{-1}
10^{-2}
on 256 fresh sequences (population; ±250 steps)
actual step s
actual step, from the exact loss change
no momentum: $-\eta g_B$ (η batch sharpness/2)
$\eta\lambda_{\max}$ / threshold, 32 fixed sequences
$\tilde{R} = s^\top \tilde{H} s / (-2 \tilde{g}^\top s)$
10^1
10^0
10^{-1}
10^{-2}
$\eta(t)/\eta_{\max}$
1.0
0.5
0.0
0
2000
4000
6000
8000
10000
step